

Engineering Survey II (GE245) Past Questions Analysis (2079 - 2082)

Unit 1: Traversing (Syllabus Hours: 8)

A. Unit Overview & Focus Topics

This is one of the most high-yielding numerical chapters. It covers basic traverse specifications, consecutive vs. independent coordinates, balancing rules (Bowditch vs. Transit), link traverses, and omitted measurements.

- **Key Repeaters:**

1. **Link Traverse Balancing (Bowditch):** Identical numerical with the same dataset and coordinates was asked in **2082 Q7** and **2079 Regular Q10(b)**.
2. **Mid-point Connection Problem:** A specific traverse problem where you connect the mid-points of CD and AB (finding length and bearing) was asked in **2081 Mid-Term Q7(b)** and **2079 Supplementary Q8(b)** with the exact same numbers.
3. **Omitted Measurements / Missing Data:** Differentiating cases and solving missing lengths/bearings is a standard 8-mark question.

B. Short Answer Questions (Recent to Older)

1. Linked Traverse & Transit Rule (2082, Q1)

How a linked traverse is balanced? How total miss-closure is balanced by Transit Rule? [5 Marks]

2. Omitted Measurements & Grid Sheet Plotting (2081, Q1)

Explain different cases of omitted measurements in theodolite traversing. Explain plotting of traverse in grid sheet by co-ordinates method. [8 Marks]

3. Coordinates & Traverse Balancing (2080, Q1)

Describe consecutive and independent coordinates. How you balance the consecutive co-ordinates? Explain the plotting method of traverse by grid coordinates. [2 + 2 + 4 = 8 Marks]

4. Closed Traverse Missing Data Numerical (2079 Regular, Q2)

For a closed traverse ABCDA, compute the missing data: **[8 Marks]**

Line	Length (m)	Bearing
AB	100.00	N 45° 30' W
BC	605.00	N 5° 30' E
CD	95.00	N 88° 20' E
DA	?	?

5. Traversing Field Measurements & Accuracy (2079 Regular, Q1)

Define closed loop and closed link traverse. Explain in brief the field measurements required in theodolite traversing with accuracy parameters (or specifications). **[8 Marks]**

6. Traverse Survey Field Procedure (2079 Supplementary, Q1)

Explain in brief field procedure of traverse survey between two known stations. **[8 Marks]**

7. Theodolite Traversing Specifications (2081 Mid-Term, Q1)

What are the field measurements necessary in theodolite traversing? Explain with appropriate specifications. **[4 Marks]**

8. Closing Error Balancing (2081 Mid-Term, Q8-b)

What is closing error? How can you balance the closing error of theodolite traverse? **[5 Marks]**

C. Long Answer Questions (Recent to Older)

1. Link Traverse Balancing - Bowditch (2082, Q7)

Balance the following coordinates and compute the total coordinates for a link traverse XABCY using Bowditch's rule. The given coordinates of X and Y are (1877.51 m N, 1290.20 m E) and (1626.50 m N, 1578.87 m E) respectively. If permissible closing error is 1:500, justify your assessment. Other observed data are as follows: **[10 Marks]**

Line	Length (m)	Bearing
XA	120.00	135° 00'
AB	111.50	119° 30'
BC	132.40	175° 00'
CY	97.60	77° 30'

(Note: This is identical to 2079 Regular Q10(b).)

2. Link Traverse with Clockwise Angles & Coordinates (2081, Q7-b)

A link traverse was run between main traverse stations "B" and "M". Clockwise angles of link traverse taken were: $ABX = 135^\circ 00' 30''$, $BXY = 98^\circ 07' 55''$, $XYM = 209^\circ 45' 02''$ and $YMN = 64^\circ 39' 33''$ respectively. Lengths of link traverse are: $BX = 31.61$ m, $XY = 22.260$ m, and $YM = 36.153$ m respectively. The coordinates of main traverse stations given are: A (42.360 m N, 18.820 m E), B (20.000 m N, 30.000 m E) and M (50.000 m N, 100.000 m E), N (70.600 m N, 65.667 m E) respectively. Calculate the final coordinates of stations X and Y. Use Bowditch method for misclosure if any. **[10 Marks]**

3. Link Traverse Balancing (2079 Regular, Q10-b)

Balance the following coordinates and compute the total coordinates for a link traverse XABCY using Bowditch's rule. The given coordinates of X and Y are (1877.51 m N, 1290.20 m E) and (1626.50 m N, 1578.87 m E) respectively. If permissible closing error is 1:500, justify your assessment. Other observed data are as follows: **[10 Marks]**

Line	Length (m)	Bearing
XA	120.00	$135^\circ 00'$
AB	111.50	$119^\circ 30'$
BC	132.40	$175^\circ 00'$
CY	97.60	$77^\circ 30'$

4. Mid-point Connection Traverse Problem (2079 Supplementary, Q8-b)

A traverse ABCDA was conducted and the following data were obtained. It was required to connect the mid-point E of CD to the mid-point F of AB. Find the length and bearing of line joining mid points E and F. **[10 Marks]**

Line	Length (m)	Bearing
AB	610.00	N $80^\circ 10'$ E
BC	510.00	N $13^\circ 00'$ E
CD	1130.00	S $80^\circ 10'$ W
DA	450.00	S $15^\circ 30'$ E

5. Mid-point Connection Traverse Problem (2081 Mid-Term, Q7-b)

A traverse ABCDA was conducted and the following data were obtained. It was required to connect the mid-point E of CD to the mid-point F of AB. Find the length and bearing of line joining mid points E and F. **[10 Marks]**

(Note: This is identical to 2079 Supplementary Q8(b).)

Unit 2: Tacheometry (Syllabus Hours: 4)

A. Unit Overview & Focus Topics

This unit deals with stadia formulas, subtense bar applications, and tangential observations.

- **Key Repeaters:**

1. **Derivation of Tacheometric Distance & RL Formula:** Differentiating and deriving formulas for a vertically held staff is asked in **2079 Regular Q3**, **2079 Supplementary Q9(b)**, and **2081 Mid-Term Q7(a)**.
2. **Subtense Bar Height/Distance Problem:** The combination of a normal staff reading and a subtense bar sight is a standard numerical pattern, seen in **2081 Q10(b)** and **2079 Regular Q4**.

B. Short Answer Questions (Recent to Older)

1. Tacheometric Reduction with Inclined Sights (2082, Q4)

A tacheometer was set up at a station A and the following readings were obtained on a staff vertically held: **[5 Marks]**

Inst. St.	Staff St.	Vertical Angle	Hair Reading (m)	Remarks
A	BM	-5° 30'	1.150, 1.800, 2.450	RL of BM = 500.50m
A	B	+8° 30'	0.750, 1.500, 2.250	

Calculate the horizontal distance between BM and B, and also the RL of B. Constant $K = 100$ and $C = 0$.

2. Tachometric Survey Radial Reductions (2080, Q2)

In order to carry out tacheometric surveying, following observations were taken through a tacheometer set up at station 'P' at height 1.235 m. **[4 + 4 = 8 Marks]**

Staff held Vertical at	Horizontal distance from P (m)	Staff Reading (m)	Angle of Elevation
Q	100	1.01	0°
R	200	2.03	0°
S	?	3.465, 2.275, 1.280	5° 24' 40"

Compute the horizontal distance of S from P and reduced level of station at S if R.L. of station P is 262.575m.

3. Subtense Bar & Stadia Tachometry (2079 Regular, Q4)

Calculate the distance and gradient between stations A and B from the given data which are observed by a tachometer from station R. Staff was vertically held at A and subtense bar at B. The subtended angle between the instrument and 2m long subtense bar was 00° 42' 15". **[8 Marks]**

Instrument Station	Sighted to	Bearing	Zenith angle	Staff readings (m)	Subtense bar height (m)
R	A	345° 00'	96° 30'	0.650, 1.250, 1.850	-
R	B	225° 00'	85° 00'	-	1.180

4. Tacheometric Distance & RL Derivation (2079 Regular, Q3)

Explain the principle of Tachometric survey and also derive the formula to determine the horizontal distance and RL of the object with respect to instrument station when the staff is held in vertical. [8 Marks]

5. Tacheometric Gradient Problem (2079 Supplementary, Q2)

A tachometric survey was done to find the gradient between X and Y. Tacheometer consist of an analytic lens was used and following observations were made from Station R on vertical staff. [8 Marks]

Inst. Station	Staff Station	Stadia Readings (m)	Vertical Angle	Bearing
R	X	0.915, 1.750, 2.585	+ 15°	345°
R	Y	0.760, 2.240, 3.715	+ 10°	75°

6. Tacheometric Bearing and Gradient (2081 Mid-Term, Q4)

Determine the gradient and bearing of PQ. The staff was held vertical. K = 100, C = 0. [4 Marks]

Instrument Station	Staff station	Bearing	Zenithal Angle	Staff Reading (m) (T, M, B)
R	P	S 60° E	79° 28'	2.36, 1.81, 1.25
R	Q	S 30° W	95° 06'	2.94, 2.12, 1.30

7. Tangential Tacheometry (2082, Q10-b)

Discuss about the tangential method of tacheometry. [2 Marks]

C. Long Answer Questions (Recent to Older)

1. Subtense Bar Theory & Computation (2081, Q10-b)

Explain subtense bar with neat sketch. How would you determine horizontal and vertical distance with the help of subtense bar? [8 Marks]

2. Tacheometric Constants & Stadia Theory (2080, Q7-a)

State the principle of stadia tacheometry and describe the field method of finding tacheometric constants. [2 + 3 = 5 Marks]

3. Tacheometric Distance & RL Derivation (2079 Supplementary, Q9-b)

Explain the principle of Tachometric survey and also derive the formula to determine the horizontal distance and RL of the object with respect to instrument station when the staff is held in vertical. [8 Marks]

4. Tacheometric Constants Stadia Theory (2081 Mid-Term, Q7-a)

State the principle of stadia tacheometry and describe the field method of finding tacheometric constants. [5 Marks]

Unit 3: Trigonometrical Levelling (Syllabus Hours: 3)

A. Unit Overview & Focus Topics

This unit is centered around calculating heights and distances when direct levelling is impossible.

- **Key Repeaters:**

1. **Inaccessible Object - Same Vertical Plane, Different Instrument Levels:** A classic derivation or numerical. Look at the complex numerical asked in **2082 Q2**, **2080 Q7(b)**, and **2079 Regular Q8(b)** where we calculate the clear height of a tower or temple using vertical/zenith angles from two instrument stations at different heights/levels.

B. Short Answer Questions (Recent to Older)

1. Inaccessible Object Same Plane Numerical (2082, Q2)

The top (Q) of a tower was sighted from two stations at very different level and in same vertical plane with Q. Find R.L of the top of tower from the following observed data: [5 Marks]

Inst. St.	H.I. (m)	R.L. (m)	Target	Zenithal Angle
P	1.87	-	Q	51° 39'
R	1.64	112.78	Q	68° 42'

The distance between instrument stations P and R is 120 m. The angle of elevation from R to 2m above the root of the staff held at P was 15° 11'.

2. Trigonometrical Levelling Methods Description (2081, Q3)

Explain the method of trigonometric leveling to determine the elevation of inaccessible object when the instrument stations and the object are in different vertical plane. [8 Marks]

3. Inaccessible Object In Same Vertical Plane (2079 Supplementary, Q3)

How can you measure the horizontal distance and elevation of an inaccessible object when the instrument positions are at very different levels; instrument stations and the elevated object are in the same vertical plane? [8 Marks]

4. Inaccessible Object Derivation (2081 Mid-Term, Q2)

Derive an expression to find the R.L. of an inaccessible object when the instrument stations are in the different vertical plane. [4 Marks]

C. Long Answer Questions (Recent to Older)

1. Double Instrument Sighting Numerical (2080, Q7-b)

The top of temple was sighted from two stations-A and B at very different level. The observed vertical angle from A and B to top of temple 'P' were $30^{\circ} 36'$ and $20^{\circ} 12'$ respectively. The vertical angle B to A to a vane at 1.5 m above the foot of the vane was $4^{\circ} 15'$. The height of instrument at A and B were 1.47 m and 1.42 m. The distance between two instrument stations was 112 m. RL of B was 1280.00 m. Find the RL of the top of the temple. Also apply the correction for refraction and curvature. [10 Marks]

2. Inaccessible Object Tower Height (2079 Regular, Q8-b)

It is required to determine the height (clear) of a Flood light tower in an arena by using a transit theodolite and for this zenith angles observation taken at 5 m and 2 m height on a target vane held on the plinth level of tower were $87^{\circ} 45'$ and $88^{\circ} 30'$ respectively. From the same instrument, zenith angle observed at top of the tower was found as $67^{\circ} 45'$. If the RL of the instrument axis was 1200.00 m, Calculate the clear height (plinth to top) of the tower. [8 Marks]

Unit 4: Contouring (Syllabus Hours: 4)

A. Unit Overview & Focus Topics

Covers contour characteristics, contour intervals, and methods of locating and interpolating contours.

- **Key Repeaters:**
 1. **Contour Interval & Horizontal Equivalent / Characteristics:** Repeated heavily across 2081 Q8(a), 2080 Q3, 2079 Regular Q10(a), and 2079 Supplementary Q10(a).
 2. **Contour Interpolation Methods:** Explaining arithmetic, graphical, and estimation methods (2081 Q9(b), 2079 Supplementary Q5).
 3. **Grid Contour Problem (5m x 5m):** A highly specific graphic interpolation numerical is repeated in 2082 Q10(a) and 2081 Mid-Term Q8(c).

B. Short Answer Questions (Recent to Older)

1. Contour Interpolation & Selection (2082, Q3)

What is meant by contour interpolation? Which method do you suggest for contour interpolation in large volume of work and how would you apply it? [5 Marks]

2. Characteristics & Selection of Contours (2080, Q3)

What are the factors that affect the selection of contour interval? Explain the various characteristics of contour line with appropriate sketch. **[3 + 5 = 8 Marks]**

3. Direct Method & Uses (2079 Regular, Q5)

Explain the direct method of contouring and write the uses of contour map. **[8 Marks]**

4. Interpolation Methods (2079 Supplementary, Q5)

What are the different methods of interpolating contours? Explain them with appropriate examples and sketch. **[8 Marks]**

5. Characteristics of Contours (2081 Mid-Term, Q3)

Discuss about the principal characteristics of contour lines with supporting neat sketches. **[4 Marks]**

6. Contour Interpolation (2081 Mid-Term, Q8-a)

What are the different methods of interpolating contours? Explain any one with appropriate examples and sketch. **[5 Marks]**

C. Long Answer Questions (Recent to Older)

1. Index Contour & Characteristics (2081, Q8-a)

What is index contour? Explain about the characteristics of contour lines with clear supporting sketches. **[8 Marks]**

2. Indirect Contouring & Uses (2081, Q9-b)

Explain the indirect method of contouring for field control. Write the uses of contour maps. **[6 Marks]**

3. Index Contour & Interpolation (2080, Q10-b)

What is index contour? Explain the method of contour interpolation. Write the uses of contour map in civil engineering works. **[10 Marks]**

4. Contour Interval & Factors (2079 Regular, Q10-a)

What is contour interval? Explain the factors that affect the contour interval. **[5 Marks]**

5. Contour Definitions & Characteristics (2079 Supplementary, Q10-a)

Describe contour interval and horizontal equivalent. Explain characteristics of contour with neat sketches. **[7 Marks]**

6. Grid Interpolation Numerical (2082, Q10-a)

If A, B, C and D are four plotted corner points of a 5m x 5m square and their ground reduced levels are 557.5, 561.5, 562.3 and 559.2m respectively. Draw the contours of 2m interval. **[2 Marks]**

7. Grid Interpolation Numerical (2081 Mid-Term, Q8-c)

If A, B, C and D are four plotted corner points of a 5m x 5m square and their ground-reduced levels are 557.5, 561.5, 562.3 and 559.2m respectively. Draw the contours of 2m interval. [5 Marks]

(Note: This is identical to 2082 Q10(a).)

Unit 5: Orientation (Syllabus Hours: 4)

A. Unit Overview & Focus Topics

This unit is highly technical, focusing on horizontal coordinate transfers through intersection and resection.

- **Key Repeaters:**

1. **Three-Point Resection (Analytical / Lehmann's Method):** Deriving and explaining coordinate calculations for three known stations. Repeated in 2082 Q6(b), 2081 Q10(a), 2079 Supplementary Q6, and 2081 Mid-Term Q6(a).
2. **Analytical Intersection Numerical:** Calculating coordinates of an unknown point B from coordinates of S and A, asked in 2079 Supplementary Q10(b).
3. **Three-Point Resection Coordinate Numerical:** A complex analytical resection coordinate calculations problem was asked in 2081 Q2 and 2079 Regular Q7(b).

B. Short Answer Questions (Recent to Older)

1. Analytical Resection (2079 Supplementary, Q6)

Explain any one method of analytical resection to calculate the position of instrument station. [8 Marks]

2. Analytical Resection (2081 Mid-Term, Q6-a)

Explain any one method of analytical resection to calculate the position of instrument station. [7 Marks]

C. Long Answer Questions (Recent to Older)

1. Analytical Resection Theory (2082, Q6-b)

Explain any one method of analytical resection to calculate the position of instrument station. [7 Marks]

2. Three-Point Resection Numerical (2081, Q2)

The following are the co-ordinates of three known station points whose directions are observed from the unknown instrument station P. [8 Marks]

Station	Easting (m) (x)	Northing (m) (y)
A	5,000	10,000
B	10,000	15,000
C	15,000	10,000

If the observed horizontal angle $APB = 45^\circ$ and $BPC = 52^\circ$, determine the coordinate of P.

3. Three-Point Resection Derivation (2080, Q9-a)

What is analytical resection? Derive an expression to find the co-ordinates of unknown points by observations to three known points. [1 + 6 = 7 Marks]

4. AC and CB Bearing/Length Intersection Numerical (2080, Q9-b)

Co-ordinate of two points A and B are given below. A third point C has been chosen in such a way that bearing of AC and CB are $29^\circ 30'$ and $45^\circ 45'$ respectively. Calculate the length of the lines AC and CB. [8 Marks]

Point	Northing	Easting
A	150	200
B	1500	1300

(Note: This is identical to 2081 Q10(a) in part b.)

5. AC and CB Bearing/Length Intersection Numerical (2081, Q10-a)

Co-ordinate of two points A and B are given below. A third point C has been chosen in such a way that bearing of AC and CB are $29^\circ 30'$ and $45^\circ 45'$ respectively. Calculate the length of the lines AC and CB. [7 Marks]

(Note: This is identical to 2080 Q9(b).)

6. Three-Point Resection Coordinate Numerical (2079 Regular, Q7-b)

Stations A, B and C have the following respective coordinates (2876.24 mE, 8754.11 mN), (3310.80 mE, 7997.25 mN) and (2959.39 mE, 7487.09 mN) respectively. Station 'O' was established and following observations were recorded by a theodolite. [10 Marks]

Pointing towards	Horizontal Circle Reading
A	$90^\circ 00' 00''$
B	$230^\circ 58' 51''$
C	$313^\circ 17' 05''$
A	$90^\circ 00' 30''$

Determine the co-ordinate of resection point 'O'.

7. Two-Station Analytical Intersection (2079 Supplementary, Q10-b)

The co-ordinates of S and A are (1309.12 m E, 1170.50 m N) and (1525.43 m E and 956.87 m N) respectively. Calculate the co-ordinates of point B which has been by intersection from stations S and A observing the following angles: $\angle BSA = 85^\circ 38' 49''$ and $\angle SAB = 55^\circ 50' 33''$. [8 Marks]

Unit 6: Curves (Syllabus Hours: 8)

A. Unit Overview & Focus Topics

This is arguably the most mathematically intensive unit of the syllabus. It covers simple circular curves, transition curves, and vertical parabolic curves.

- **Key Repeaters:**

1. **Vertical Parabolic Curve Table Design:** A road alignment parabolic curve design question with the exact same numerical data (IP RL = 1000m, IP chainage = 1500m, grade -4.5% followed by +3.5%, L=200m, peg interval=20m) was repeated in **2081 Q6**, **2079 Regular Q6**, and **2079 Supplementary Q7(b)**! Differentiating with slightly different coordinates (grade -3.5% and upgrade +4.5%, IP RL = 900m, IP chainage = 2450m, L=180m, interval=20m) was asked in **2082 Q8(b)** and **2080 Q6**.
2. **Rankine's Deflection Method:** Standard step-by-step curve tables for Rankine's method are repeated in **2081 Q8(b)** and **2079 Regular Q9(a)** (intersection angle = 145°, IP chainage = 1580m, degree = 5°, LC = 10", peg interval = 30m) and **2081 Mid-Term Q6(b)**.
3. **Derivation of Tangential Deflection Angle:** Deriving $\delta = (90^\circ \times c)/(\pi R)$ was asked in **2082 Q5** and **2079 Supplementary Q4**.
4. **Derivation of Transition Curve Deflection Angle:** Deriving $\delta = 573L^2 / RL$ was asked in **2081 Q9(a)**.

B. Short Answer Questions (Recent to Older)

1. Tangential Deflection Formula Derivation (2082, Q5)

Derive the formula of Tangential angle, $\delta = (90^\circ \times c)/(\pi R)$ and deflection angle $\Delta_n = \delta_1 + \delta_2 + \delta_3 + \dots + \delta_n$. **[5 Marks]**

2. Vertical Curve Design (2081, Q6)

Prepare a table giving all necessary data for setting out a vertical curve. In a road alignment a grade of - 4.5% followed by + 3.5%, R.L. of I.P. = 1000 m, chainage of IP = 1500 m. Connect the two grades by a parabolic curve 200 m long. Take peg interval = 20 m. **[8 Marks]**

(Note: This is identical to 2079 Regular Q6.)

3. Vertical Parabolic Curve Design (2080, Q6)

A down grade of 3.5% is followed by an upgrade of 4.5%. The reduced level and chainage of IP are 900.00m and 2450.00 m respectively. A vertical parabolic curve 180 m long is to be introduced to connect the two grades. The pegs are to-be-fixed-at 20 m intervals. Calculate the RLS of curve points including lowest point. **[8 Marks]**

4. Vertical Curve Design (2079 Regular, Q6)

Prepare a table giving all necessary data for setting out a vertical curve. In a road alignment a grade of - 4.5% followed by + 3.5%, R.L. of I.P. = 1000 m, chainage of IP = 1500 m. Connect the two grades by a parabolic curve 200 m long. Take peg interval = 20 m. **[8 Marks]**

5. Tangential Deflection Formula Derivation (2079 Supplementary, Q4)

Explain degree of curve with neat sketch. Derive the formula of Tangential angle, $\delta = (90^\circ \times c)/(\pi R)$ and deflection angle $\Delta_n = \delta_1 + \delta_2 + \delta_3 + \dots + \delta_n$. **[8 Marks]**

6. Degree of Curve Definitions (2081 Mid-Term, Q5)

What is degree of curve? Describe the elements of simple circular curve. **[4 Marks]**

7. Simple Circular Curve Elements (2082, Q9-a)

Find the elements of simple circular curve. **[3 Marks]**

C. Long Answer Questions (Recent to Older)

1. Vertical Parabolic Curve Design (2082, Q8-b)

A down grade of 3.5% is followed by an upgrade of 4.5%. The reduced level and chainage of IP are 900.00m and 2450.00m respectively. A vertical parabolic curve 180m long is to be introduced to connect the two grades. The pegs are to be fixed at 20m intervals. Calculate the RLS of curve points including lowest point. **[7 Marks]**

(Note: This is identical to 2080 Q6.)

2. Transition Curve Derivation (2081, Q9-a)

Derive the formula for deflection angle (δ) in transition curve, $\delta = 573L^2/(R L_s)$ minutes. **[4 Marks]**

3. Setting out by Offsets from Long Chord (2081, Q11-c)

Explain the setting out of simple circular curve by offsets from long chord. **[5 Marks]**

4. Rankine's Deflection Method Table (2081, Q8-b)

Compute the data for setting out a simple circular curve by Rankine's deflection angle method from the following information: **[7 Marks]**

- Angle of intersection = $145^\circ 00'$
- Chainage of point of intersection = 1580 m
- Degree of curve = 5°
- Least count of the theodolite = 10"
- Peg interval = 30 m

5. Circular Curve and Transition Curve Layout (2080, Q8)

- a) What is degree of curve? Describe the elements of simple circular curve. **[2 + 3 = 5 Marks]**

- b) A road 8 m wide is to deflect through an angle of 60° with the Centre line radius of 300 m, the chainage of intersection point being 3605.00 m. A transition curve is to be used at each end of the circular curve of such a length that the rate of gain of radial acceleration is 0.5 m/s^3 , when the speed is 50 km/hr. Find out: **[2 + 2 + 2 + 4 = 10 Marks]**

Marks]

- i. Length of the transition curve.
- ii. Super elevation.
- iii. Chainage of all junction points.
- Lay out the transition curve by the method of deflection angles, taking peg interval as 5 m.

6. Rankine's Deflection Method Table (2079 Regular, Q9-a)

Compute the data for setting out a simple circular curve by Rankine's deflection angle method from the following information: **[7 Marks]**

- Angle of intersection = 145°
- Chainage of point of intersection = 1580 m
- Degree of curve = 5°
- Least count of the theodolite = 10"
- Peg interval = 30 m

7. Vertical Parabolic Curve Table (2079 Supplementary, Q7-b)

Prepare a table giving all necessary data for setting out a vertical curve. In a road alignment a grade of - 4.5% followed by + 3.5% R.L. of I.P. = 1000 m, chainage of IP = 1500 m connect the two grades by a parabolic curve 200 m long. Take peg interval = 20 m. **[10 Marks]**

8. Rankine's Deflection Method (2081 Mid-Term, Q6-b)

Compute the data for setting out a simple circular curve by Rankine's deflection angle method from the following information: **[8 Marks]**

- Angle of intersection = 145°
- Chainage of point of intersection = 1580 m
- Degree of curve = 5° (Note: written as "50" in some versions, but standard data is 5°)
- Least count of the theodolite = 10"
- Peg interval = 30 m

Unit 7: Hydrographic Survey (Syllabus Hours: 2)

A. Unit Overview & Focus Topics

Focuses on depth measurements, soundings, and cross-section measurements of water bodies.

- **Key Repeaters:**

1. **Locating Sounding Positions:** *Differentiating methods of locating soundings (like from boat, shore, intersecting lines) is asked in 2081 Q5, 2080 Q5, 2079 Regular Q9(b), and 2079 Supplementary Q9(a).*

B. Short Answer Questions (Recent to Older)

1. Hydrographic Survey & Soundings (2081, Q5)

Write down the purpose of hydrographic survey & explain the method of locating sounding positions. **[8 Marks]**

2. Sounding Positions (2080, Q5)

How can you determine the depth of large water body? Explain the method of locating sounding positions. **[3 + 5 = 8 Marks]**

C. Long Answer Questions (Recent to Older)

1. Stream Cross-Section Measurement (2082, Q8-a)

How would you measure the cross-section of a small stream? **[3 Marks]**

2. Soundings & Equipment (2079 Supplementary, Q9-a)

Define sounding, and state the equipment and methods used for locating and making soundings. **[7 Marks]**

3. Sounding Point Location Methods (2079 Regular, Q9-b)

Define vertical and horizontal control. Explain the methods of locating sounding points in hydrographic surveying. **[8 Marks]**

Unit 8: Photogrammetry and Remote Sensing (Syllabus Hours: 4)

A. Unit Overview & Focus Topics

Covers definitions of aerial photogrammetry, scale of a photograph, relief displacement, and remote sensing.

- **Key Repeaters:**

1. **Relief Displacement Derivation:** *Standard analytical derivation with sketch, asked in 2081 Q9(c), 2079 Regular Q7(a), and 2079 Supplementary Q11(b).*
2. **Active vs. Passive Remote Sensing:** *Asked in 2082 Q9(b) and 2081 Q11(b).*
3. **Scale of Vertical Photograph:** *Questions on deriving or using scale equations (2080 Q10(a), 2079 Supplementary Q8(a)).*

B. Short Answer Questions (Recent to Older)

1. Relief Displacement Derivation (2079 Regular, Q7-a)

With the help of neat sketch explain with expression the relief displacement of a vertical photograph. [5 Marks]

2. Scale of Vertical Photograph Derivation (2079 Supplementary, Q8-a)

Determine the expression for the scale of vertical photograph. [5 Marks]

C. Long Answer Questions (Recent to Older)

1. Photogrammetric Scale Calculation (2080, Q10-a)

A line 2350 m long laying at an elevation of 500 m measures 10.6 cm on vertical photograph. The focal length of camera used is 20 cm. Determine the scale of the photograph for an area having an elevation of 1200 m. [5 Marks]

2. Relief Displacement Scale (2081, Q9-c)

Determine the expression for the relief displacement of a vertical photograph. [5 Marks]

3. Remote Sensing & Applications (2081, Q11-b)

What is remote sensing? Explain types of remote sensing and applications of remote sensing. [5 Marks]

4. Active vs. Passive Remote Sensing (2082, Q9-b)

Differentiate between the active and passive remote sensing. [4 Marks]

Unit 9: Field Astronomy and GPS System (Syllabus Hours: 4)

A. Unit Overview & Focus Topics

This unit focuses on the fundamentals of the Global Positioning System (GPS) and coordinate systems.

- **Key Repeaters:**

1. **GPS Components & Working Principle:** Differentiating segments (Space, Control, User) and how trilateration works. Repeated heavily in **2082 Q10(c)**, **2081 Q11(a)**, **2080 Q11(a)**, **2079 Regular Q8(a)**, and **2079 Supplementary Q7(a)**.

B. Short Answer Questions (Recent to Older)

1. ***GPS Components & Working Principle (2079 Supplementary, Q7-a)***

What is GPS? Write the components and working principle of GPS. **[5 Marks]**

C. Long Answer Questions (Recent to Older)

1. ***GPS Components & Engineering Applications (2082, Q10-c)***

Explain the function of GPS components. Write down the engineering applications of GPS. **[3 + 3 = 6 Marks]**

2. ***GPS Components & Applications (2081, Q11-a)***

What is GPS? Write the components and applications of GPS. **[5 Marks]**

3. ***GPS Components & Working Principle (2079 Regular, Q8-a)***

What is GPS? Write the components and working principle of GPS. **[7 Marks]**

Unit 10: Geographic Information Systems (GIS) & Short Notes

(Note: While GIS is closely associated with Unit 9 in practice, the questions on GIS are typically framed as Short Notes or Applications in civil engineering).

A. Key Repeaters

- **GIS Components and Applications:** Differentiating hardware, software, data, people, and methods, and how it is applied to road, water, or municipal civil projects. Repeated across **2082 Q9(c), 2081 Q11(d), 2080 Q11(d), 2079 Regular Q11(a), and 2079 Supplementary Q11(a).**

B. Long Answer Questions & Short Notes

1. GIS Applications (2082, Q9-c)

What are the engineering applications of GIS? **[3 Marks]**

2. Short Note: GIS Components & Applications (2081, Q11-d)

Components of GIS and applications of GIS in Civil engineering. **[5 Marks]**

3. Short Note: GIS Components & Applications (2080, Q11-d)

Components of GIS and applications of GIS in Civil engineering. **[5 Marks]**

4. Short Note: GIS Components & Applications (2079 Regular, Q11-a)

Component of GIS and its application. **[5 Marks]**

5. Short Note: GIS Components & Applications (2079 Supplementary, Q11-a)

Components of GIS and its application. **[5 Marks]**

6. Short Note: GIS Applications (2079 Supplementary, Q11-d)

Application of GIS in Civil Engineering. **[5 Marks]**

7. Short Note: Remote Sensing Applications (2079 Regular, Q11-b)

Application of remote sensing in civil engineering. **[5 Marks]**

Part 3: Chapter-Wise Prioritization Table & Planner

Based on the statistical weightage of marks and frequency of occurrence in the past 4 years of exam papers, the 10 units are ranked below from **Highest Priority** to **Lowest Priority** to guide your preparation.

Rank	Syllabus Unit	Estimated Exam Weightage (%)	Focus Priority	Primary Areas of Focus (High-Yield Topics)
1	Unit 6: Curves	20% - 25%	Critical (Highest)	Parabolic vertical curve table design, Rankine's deflection angle calculations, and Transition deflection angle derivations.
2	Unit 1: Traversing	18% - 22%	Critical (Highest)	Bowditch link traverse coordinate balancing, consecutive vs. independent coordinate concepts, and omitted measurements.
3	Unit 2: Tacheometry	15% - 18%	High Focus	Stadia formula derivations (vertically held staff), subtense bar height distance reductions, and vertical angle gradients.
4	Unit 3: Trigonometrical Levelling	12% - 15%	High Focus	R.L. reductions for double instrument setup sightings (same and different vertical planes), and curvature/refraction corrections.
5	Unit 4: Contouring	10% - 12%	High Focus	Factors affecting contour intervals, characteristics of contours with sketches, and direct vs. indirect contouring.
6	Unit 5: Orientation	8% - 12%	Medium Focus	Analytical resection derivations, Lehman's three-point resection method, and coordinates from angular sightings.
7	Unit 8: Photogrammetry	6% - 10%	Medium Focus	Expression for scale of vertical photographs, and relief displacement derivations/sketches.
8	Unit 9: GPS & Astronomy	5% - 8%	Medium Focus	GPS components (Space, Control, User segments), trilateration principle, and applications.
9	Unit 10 (or GIS Short Notes)	5% - 7%	Low Focus	Standard GIS definition, core components, and spatial data applications in Civil Engineering.
10	Unit 7: Hydrographic Survey	3% - 5%	Low Focus	Purpose of sounding, equipment, and methods of locating sounding points.

1. **The 60% Rule:** Units 6, 1, and 2 make up over **60% of the total marks** on average. Master these three chapters (especially the numerical tables for vertical curves, Bowditch traverses, and subtense/stadia observations) first.